

SHUMEN, Bulgaria

WMO: 155440

Lat: 43.267N

Lon: 26.933E

Elev: 220

StdP: 98.71

Time Zone: 2.00 (EUE)

Period: 82-90

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.0	-15.2	1.0	-10.8	-13.0	1.3	-7.9	15.8	7.4	13.7	2.8	1.4	N/A	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.5	20.2	29.8	19.5	28.1	18.7	21.4	29.0	20.4	27.4	19.7	26.2	3.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.2	24.4	18.0	13.3	22.9	17.2	12.6	22.3	63.2	28.9	59.4	27.3	57.0	26.4	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
12.2	9.9	8.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-14.9	34.8	3.2	2.4	-17.2	36.5	-19.1	37.9	-20.9	39.2	-23.3	40.9
			-15.3	23.9	3.2	2.1	-17.6	25.3	-19.4	26.5	-21.2	27.7	-23.5	29.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	0.6	0.9	5.0	11.0	15.8	19.0	21.3	21.1	18.1	11.5	4.9	2.3		
	DBStd	8.64	4.59	5.41	5.54	3.71	3.42	2.56	2.76	2.56	2.92	3.99	4.56	4.56		
	HDD10.0	1186	293	257	171	31	2	0	0	0	0	29	162	241		
	HDD18.3	2951	551	488	412	220	91	22	4	7	39	215	404	497		
	CDD10.0	1557	1	2	17	62	183	271	349	343	243	75	8	3		
	CDD18.3	281	0	0	0	1	14	43	96	92	32	3	0	0		
	CDH23.3	2517	0	0	0	26	137	357	857	804	307	29	0	0		
	CDH26.7	720	0	0	0	5	17	74	298	247	71	8	0	0		
(14) Wind	WSAvg	2.3	2.3	2.6	2.6	2.7	2.3	2.7	2.3	2.1	1.8	1.9	2.1	2.4		
(15) Precipitation	PrecAvg	559	36	39	39	51	65	76	50	42	36	37	48	49		
	PrecMax	901	96	103	108	95	162	164	108	129	104	119	174	221		
	PrecMin	351	2	11	4	10	6	20	11	1	1	2	4	3		
	PrecStd	115	26	23	27	21	42	36	31	29	28	27	34	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	15.7	22.3	26.4	28.6	31.7	35.8	32.7	31.3	26.9	18.1	17.1		
		MCWB	8.2	9.7	11.9	14.4	17.9	21.3	23.2	20.1	17.8	17.5	12.6	10.5		
	2%	DB	11.5	13.4	19.0	22.8	26.3	28.5	32.3	30.8	28.6	22.6	15.7	13.4		
		MCWB	7.3	8.4	10.7	13.4	16.6	19.4	21.7	19.5	17.6	14.8	11.7	9.0		
	5%	DB	9.2	11.0	16.5	20.7	24.8	27.0	29.8	29.5	26.7	20.2	13.6	10.9		
		MCWB	6.2	7.2	9.9	12.6	15.9	18.8	20.2	19.0	17.3	14.1	10.3	7.4		
	10%	DB	6.8	8.2	14.1	18.4	23.1	25.4	27.8	27.9	24.8	18.0	11.5	8.8		
		MCWB	4.5	5.3	8.6	11.5	15.3	18.0	19.2	18.5	16.7	13.0	9.2	6.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	10.3	12.8	15.7	19.2	21.6	24.1	21.3	20.0	17.7	13.2	11.0		
		MCDB	12.5	14.1	19.4	23.5	26.7	29.1	33.3	29.5	24.9	26.1	17.1	14.8		
	2%	WB	7.6	9.0	11.6	14.4	17.8	20.4	22.4	20.5	19.0	15.7	12.1	9.6		
		MCDB	10.9	13.2	17.9	21.5	24.1	27.4	30.5	28.3	26.0	20.4	15.5	12.6		
	5%	WB	6.2	7.5	10.2	13.0	16.9	19.6	20.9	19.9	18.2	14.6	10.7	7.9		
		MCDB	9.2	10.3	16.2	19.7	23.0	25.7	27.7	27.3	24.7	19.2	12.8	10.8		
	10%	WB	4.6	5.6	8.7	11.9	15.9	18.7	19.9	19.3	17.3	13.5	9.5	6.2		
		MCDB	6.5	8.2	13.6	17.9	21.4	24.1	26.3	26.1	23.1	17.5	11.5	8.5		
(35) Mean Daily Temperature Range	MDBR		6.9	7.1	8.3	10.2	11.1	10.3	11.0	11.5	11.8	9.6	7.4	6.5		
	5% DB	MCDBR	10.7	11.2	13.5	15.1	13.8	13.3	14.5	14.3	15.0	13.6	11.0	10.9		
		MCWBR	7.5	7.2	7.1	7.2	5.9	5.8	5.4	5.3	5.8	6.4	7.0	6.8		
	5% WB	MCDBR	10.4	9.8	13.2	13.8	12.9	12.3	12.8	12.9	13.1	12.1	9.5	9.8		
		MCWBR	7.5	6.9	7.2	7.1	6.3	5.7	5.7	5.4	5.5	5.8	6.4	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.364	0.401	0.449	0.425	0.429	0.438	0.448	0.421	0.385	0.358	0.336		
	taud		2.334	2.262	2.201	2.075	2.218	2.239	2.227	2.194	2.242	2.328	2.352	2.344		
	Ebn at Noon		782	816	830	817	850	845	833	810	802	784	746	741		
	Edh at Noon		83	106	127	156	139	137	137	137	121	98	82	76		
(44) All-Sky Solar Radiation	RadAvg		1.50	2.24	3.35	4.68	5.85	6.41	6.52	5.86	4.26	2.73	1.65	1.27		
	RadStd		0.21	0.29	0.31	0.52	0.43	0.48	0.44	0.31	0.35	0.28	0.21	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	-108	+81	+50

Nomenclature: See separate page